

To understand the complexity of the gradient of f , we need some preparation. For an $n \times n$ matrix M , recall the definition of the operator norm:

$$\|M\| := \max\{\|Mz\| : z \in \mathbb{R}^n, \|z\| = 1\},$$

where $\|z\|$ denotes the Euclidean norm of z . Equivalently, $\|M\|$ is the largest singular value of M . For $x \in \mathcal{P}_n$, let $M(x)$ be the symmetric matrix whose (i, j) th entry is x_{ij} , with the convention that $x_{ij} = x_{ji}$ and $x_{ii} = 0$. Define the operator norm of x as

$$\|x\|_{\text{op}} := \|M(x)\|.$$

The following lemma estimates the entropy of the unit cube under this norm.

Lemma 8.3 *For any $\tau \in (0, 1)$, there is a finite set of $n \times n$ matrices $\mathcal{W}(\tau)$ such that*

$$|\mathcal{W}(\tau)| \leq e^{34(n/\tau^2) \log(51/\tau^2)},$$

and for any $n \times n$ matrix M with entries in $[0, 1]$, there exists $W \in \mathcal{W}(\tau)$ such that

$$\|M - W\| \leq n\tau.$$

In particular, for any $x \in \mathcal{P}_n$ there exists $W \in \mathcal{W}(\tau)$ such that

$$\|M(x) - W\| \leq n\tau.$$

Proof Suppose that $n < 17/\tau^2$. Let $\mathcal{W}(\tau)$ consist of all $n \times n$ matrices whose entries are all integer multiples of τ and belong to the interval $[0, 1]$. It is then easy to see that for any M with entries in $[0, 1]$, there exists $W \in \mathcal{W}(\tau)$ such that $\|M - W\| \leq n\tau$. Moreover, since $n < 17/\tau^2$,

$$|\mathcal{W}(\tau)| \leq \tau^{-n^2} \leq e^{9(n/\tau^2) \log(1/\tau^2)},$$

completing the proof in this case.

Next, suppose that $n \geq 17/\tau^2$. Let l be the integer part of $17/\tau^2$ and $\delta = 1/l$. Let $\mathcal{A}$ be a finite subset of the unit ball of $\mathbb{R}^n$ such that any vector inside the ball is at Euclidean distance $\leq \delta$ from some element of $\mathcal{A}$. (In other words, $\mathcal{A}$ is a δ -net of the unit ball under the Euclidean metric.) The set $\mathcal{A}$ may be defined as a maximal set of points in the unit ball such that any two are at a distance greater than δ from each other. Since the balls of radius $\delta/2$ around these points are disjoint and their union is contained in the ball of radius $1 + \delta/2$ centered at zero, it follows that $|\mathcal{A}|C(\delta/2)^n \leq C(1 + \delta/2)^n$, where C is the volume of the unit ball. Therefore,

$$|\mathcal{A}| \leq (3/\delta)^n. \tag{8.2.4}$$

Take any $x \in \mathcal{P}_n$. Suppose that M has singular value decomposition

$$M = \sum_{i=1}^n \lambda_i u_i v_i^t,$$

where $\lambda_1 \geq \lambda_2 \geq \dots \lambda_n \geq 0$ are the singular values of M , and $u_1, \dots, u_n$ and $v_1, \dots, v_n$ are singular vectors, and v_i^t denotes the transpose of the column vector v_i . Assume that the u_i 's and v_i 's are orthonormal systems. Since the elements of M all belong to the interval $[0, 1]$, it is easy to see that $\lambda_1 \leq n$ and $\sum \lambda_i^2 \leq n^2$. Due to the second inequality, there exists $y \in \mathcal{A}$ such that

$$\sum_{i=1}^n (n^{-1} \lambda_i - y_i)^2 \leq \delta^2. \quad (8.2.5)$$

Let $z_1, \dots, z_n$ and $w_1, \dots, w_n$ be elements of $\mathcal{A}$ such that for each i ,

$$\sum_{j=1}^n (u_{ij} - z_{ij})^2 \leq \delta^2 \quad \text{and} \quad \sum_{j=1}^n (v_{ij} - w_{ij})^2 \leq \delta^2, \quad (8.2.6)$$

where u_{ij} denotes the j th component of the vector u_i , etc. Define two matrices V and W as

$$V := \sum_{i=1}^{l-1} \lambda_i u_i v_i^t \quad \text{and} \quad W := \sum_{i=1}^{l-1} n y_i z_i w_i^t.$$

Note that since $\sum \lambda_i^2 \leq n^2$ and λ_i decreases with i , therefore for each i , $\lambda_i^2 \leq n^2/i$. Thus,

$$\begin{aligned} \|M - W\| &\leq \|M - V\| + \|V - W\| \\ &\leq \frac{n}{\sqrt{l}} + \|V - W\|. \end{aligned}$$

Next, note that by (8.2.6), the operator norms of the rank-one matrices $(u_i - z_i)v_i^t$ and $z_i(v_i - w_i)^t$ are bounded by δ . And by (8.2.5), $|\lambda_i - n y_i| \leq n\delta$ for each i . Therefore

$$\begin{aligned} \|V - W\| &\leq \left\| \sum_{i=1}^{l-1} (\lambda_i - n y_i) u_i v_i^t \right\| + \left\| \sum_{i=1}^{l-1} n y_i (u_i - z_i) v_i^t \right\| \\ &\quad + \left\| \sum_{i=1}^{l-1} n y_i z_i (v_i - w_i)^t \right\| \\ &\leq \max_{1 \leq i \leq l-1} |\lambda_i - n y_i| + 2 \sum_{i=1}^{l-1} n |y_i| \delta \\ &\leq n\delta + 2n\delta \left((l-1) \sum_{i=1}^{l-1} y_i^2 \right)^{1/2} \leq n\delta + 2n\delta \sqrt{l-1} \leq \frac{3n}{\sqrt{l}}. \end{aligned}$$

Thus,

$$\|M - W\| \leq \frac{4n}{\sqrt{l}} \leq \frac{4n}{\sqrt{\frac{17}{\tau^2} - 1}} \leq \frac{4n}{\sqrt{\frac{16}{\tau^2}}} = n\tau.$$

Let $\mathcal{W}(\tau)$ be the set of all possible W 's constructed in the above manner. Then $\mathcal{W}(\tau)$ has the required property, and by (8.2.4),

$$|\mathcal{W}(\tau)| \leq \text{The number of ways of choosing}$$

$$\begin{aligned} & y, z_1, \dots, z_{l-1}, w_1, \dots, w_{l-1} \in \mathcal{A} \\ &= |\mathcal{A}|^{2l-1} \leq (3/\delta)^{2nl} = e^{2nl \log(3l)}. \end{aligned}$$

This completes the proof of the lemma. $\square$

Let r be a positive integer. Let K_r be the complete graph on the vertex set $\{1, \dots, r\}$. For any set of edges A of K_r , any $q = (q_1, \dots, q_r) \in [n]^r$, and any $x \in \mathcal{P}_n$, let

$$P(x, q, A) := \prod_{\{a,b\} \in A} x_{q_a q_b},$$

with the usual convention that the empty product is 1. Note that if $q_a = q_b$ for some $\{a, b\} \in A$, the $P(x, q, A) = 0$ due to our convention that $x_{ii} = 0$ for each i . Next, note that if A and B are disjoint sets of edges, then

$$P(x, q, A \cup B) = P(x, q, A)P(x, q, B). \quad (8.2.7)$$

Lemma 8.4 *Let A and B be sets of edges of K_r , and let $e = \{\alpha, \beta\}$ be an edge that is neither in A nor in B . Then for any $x, y \in \mathcal{P}_n$,*

$$\left| \sum_{q \in [n]^r} P(x, q, A)P(y, q, B)(x_{q_\alpha q_\beta} - y_{q_\alpha q_\beta}) \right| \leq n^{r-1} \|x - y\|_{\text{op}}.$$

Proof By relabeling the vertices of K_r and redefining A and B , we may assume that $\alpha = 1$ and $\beta = 2$.

Let A_1 be the set of all edges in A that are incident to 1. Let A_2 be the set of all edges in A that are incident to 2. Note that since $\{1, 2\} \notin A$, therefore A_1 and A_2 must be disjoint. Similarly, let B_1 be the set of all edges in B that are incident to 1 and let B_2 be the set of all edges in B that are incident to 2. Let $A_3 = A \setminus (A_1 \cup A_2)$ and $B_3 = B \setminus (B_1 \cup B_2)$. By (8.2.7),

$$P(x, q, A) = P(x, q, A_1)P(x, q, A_2)P(x, q, A_3)$$